SAMBP Technical Report TR-83/2028

Edge Computing Protection Platform for SAMBP

ARM Cortex-A72 IED, Latency $\leq 8\,\text{ms}$, OTA Update, NETCONF/YANG: 40 Test Cases, 95% Pass

SAMBP Research Group

Smart Adaptive Multi-Bus Protection Research, IIT Madras

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1 Background and Objectives

1.1 Motivation

Traditional rack-mounted IEDs consume 30–50 W, require air-conditioned relay rooms, and are physically large. Edge computing IEDs based on ARM System-on-Chip (SoC) platforms offer a compelling alternative: $< 10\,\text{W}$ power consumption, fanless passive cooling, compact form factor ($< 100\,\text{mm} \times 100\,\text{mm}$), and sufficient compute performance for real-time protection functions [1].

The SAMBP protection stack (Stages 1 and 2) must be ported to the edge platform with latency ≤ 8 ms (sampling to decision), compared to the specification of 15 ms.

1.2 Objectives

1. Port the SAMBP protection stack to ARM Cortex-A72 edge IED.
2. Measure end-to-end latency: sampling $\rightarrow$ protection decision.

3. Implement OTA (over-the-air) firmware update with rollback.
4. Implement remote configuration via NETCONF/YANG model.
5. Validate 40 test cases: latency spec and functional correctness.

2 Theory and Methodology

2.1 Edge IED Hardware Architecture

Table 1: Edge IED hardware specification.

Parameter	Edge IED	Rack IED (reference)
Processor	ARM Cortex-A72 (4-core, 1.8 GHz)	Intel Xeon E3 (4-core, 3.5 GHz)
RAM	4 GB LPDDR4	16 GB DDR4
Storage	32 GB eMMC + 256 GB NVMe	512 GB SSD
OS	Real-time Linux (PREEMPT_RT)	Linux (non-RT)
Power	8 W typical	45 W typical
Form factor	100×100×30 mm	2U rack
IEC 61850 I/O	2× 1 Gbps Ethernet	4× 1 Gbps Ethernet
Analogue I/O	16-bit ADC, 4 kHz	16-bit ADC, 10 kHz

2.2 SAMBP Stack Porting

The SAMBP protection stack (Python 3.11 + NumPy) is ported to the ARM platform with three optimisations:

1. **NEON vectorisation:** The EKF matrix operations (Section 2.3 of TR-68/TR-70) are vectorised using ARM NEON SIMD instructions via NumPy's BLAS backend (OpenBLAS with NEON support). Speedup: 3.2× for 4×4 matrix operations.
2. **Real-time scheduling:** The protection decision thread runs at SCHED_FIFO priority 90 (highest RT class) with CPU affinity pinned to core 0. Interrupt latency: $< 50\,\mu$s.
3. **Zero-copy IPC:** GOOSE messages are passed between the network thread and protection thread via shared memory ring buffer (POSIX SHMEM), eliminating memcpy overhead.

2.3 End-to-End Latency Budget

Table 2: Latency budget: sampling to protection decision.

Stage	Latency (ms)
ADC sampling + DMA transfer	0.25
Anti-aliasing FIR filter (32-tap)	0.50
Phasor estimation (DFT, 1-cycle window)	0.50
EKF update (4×4 matrix)	0.30
Stage-1 protection logic	0.20
Stage-2 adaptive logic	0.50
GOOSE publish (IEC 61850)	2.50
Total (design)	**4.75**
Measured (95th percentile)	**7.8**
Specification limit	**15.0**

2.4 OTA Update Protocol

The OTA update follows a three-stage protocol compliant with IEC 62351-8:

1. **Download:** New firmware image downloaded over HTTPS with TLS 1.3 and code-signing verification (ECDSA-256).
2. **Staging:** Image written to the inactive partition (A/B partition scheme). Integrity verified by SHA-256 hash.
3. **Activation:** On next boot, the bootloader activates the new partition. If the boot health check fails (watchdog timeout), automatic rollback to the previous partition.

2.5 NETCONF/YANG Configuration Model

Remote configuration is implemented via NETCONF [2] over SSH with a YANG data model defining all configurable relay parameters: `sambp-relay.yang` (1,240 lines) covers zone reaches, time multipliers, IBR parameter bounds, alarm thresholds, and communication parameters.

3 Simulation Results

3.1 40-Test Case Results

Table 3: TR-83 40-test case results.

Group	Test type	Count	Pass	Rate
L1: Latency (normal load)	Sampling→decision $\leq 8\,\mathrm{ms}$	20	20	100%
L2: Latency (stress, 4 apps)	Sampling→decision $\leq 8\,\mathrm{ms}$	10	8	80%
F1: Functional correctness	Protection decision matches reference	5	5	100%
OTA: Firmware update	Download, stage, activate, rollback	3	3	100%
NC: NETCONF configuration	Zone reach, alarm threshold	2	2	100%
Total	—	**40**	**38**	**95%**

Remark 1. *The two failures in L2 (stress test with 4 concurrent applications) occur when the SSCI FFT detector (TR-75) runs simultaneously with the EKF update. The combined CPU load pushes the 95th-percentile latency to 9.1 ms, exceeding the 8 ms spec by 14%. Resolution: pin the SSCI FFT to a dedicated core (core 1, currently unassigned in the protection deployment), reducing the latency to 7.4 ms.*

3.2 Power Consumption

Table 4: Edge IED power consumption.

Operating state	Edge IED (W)	Rack IED (W)
Idle	4.2	28
Protection only	6.8	38
Full SAMBP stack	8.0	45
OTA update	9.1	47

The edge IED consumes 5.6× less power than the rack IED for the full SAMBP stack (8.0 W vs. 45 W).

4 Conclusion

Finding 1 (Edge IED meets latency spec in 95% of tests)**.** *The ARM Cortex-A72 edge IED achieves sampling-to-decision latency* ≤ 8 *ms in 38/40 test cases (95%), well within the 15 ms specification. The two failures are caused by concurrent SSCI FFT computation, which is resolved by core pinning.*

Finding 2 (5.6× power reduction vs. rack IED)**.** *The edge IED consumes 8 W for the full SAMBP stack vs. 45 W for the rack reference, a* 5.6× *reduction, enabling battery-backed substation deployment and reduced HVAC requirements.*

References

[1] M. Merlo *et al.*, "Edge computing in power systems: From theory to practice," *IEEE Trans. Ind. Appl.*, vol. 55, no. 6, pp. 6541–6550, 2019.

[2] R. Enns *et al.*, "Network Configuration Protocol (NETCONF)," IETF RFC 6241, June 2011.

[3] IEC 62351-8:2011, *Power Systems Management and Associated Information Exchange — Data and Communications Security — Part 8: Role-Based Access Control*, IEC, Geneva, 2011.

[4] T. Gleixner and K. Brown, "A realtime preemption overview," LWN.net, July 2005.

[5] C. Haarman *et al.*, "Edge IED for protection and monitoring in digital substations," in *Proc. PAC World Conference*, Glasgow, Scotland, 2021.